

broadcast

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STEP 1: Coarse Search | 3 AVX-2 Instructions

__m256 comparison:

Group 0

bin 0
2.5



$x \geq 2.5$

Group 1

bin 8
5.1



$x \geq 5.1$

Group 2

bin 16
7.8



$x \geq 7.8$

Group 3

bin 24
10.2



$x \geq 10.2$

Group 4

bin 32
12.7



$x < 12.7$

Group 5

bin 40
15.3



$x < 15.3$

Group 6

bin 48
18.0



$x < 18.0$

Group 7

bin 56
20.5



$x < 20.5$

movemask_ps:

mask = 0b00001111 (15)

popcnt:

K = 4

← Group 4 selected!

STEP 2: Fine Search within Group 3 | 4 AVX-2 Instructions

__m256 comparison:

bin 24

10.2



bin 25

10.9



bin 26

11.4



bin 27

11.9



bin 28

12.1



bin 29

12.2



bin 30

12.4



bin 31

12.6



movemask_ps:

mask = 0b00000111 (7)

popcnt:

cnt = 3

Final Index:

$(K \ll 3) + \text{cnt} - 1 = (4 \times 8) + 3 - 1 = 26$